Interval certification of the smooth-arc curvature discrepancy

Computational supplement to manuscript V423

23 September 2026

Result. The value already printed in V423 is correctly rounded to 24 decimal places:

$$\boxed{D_\kappa \approx 0.20378375953604944382608 \quad \text{(correctly rounded to 24 decimal places).}}$$

This is a correctly rounded approximation, not an exact finite decimal representation. The following rigorous enclosure is deliberately widened from the full computational output:

$$\begin{aligned} 0.20378375953604944382608062942260682 8 &< D_\kappa \\ &< 0.20378375953604944382608062942260682 9. \end{aligned} \tag{1}$$

Both bounds lie strictly inside the rounding cell

$$\big(0.2037837595360494438260805,\ 0.2037837595360494438260815\big).$$

Thus the decimal retained in V423 is certified without changing any of its 24 digits. This certificate concerns the smooth-arc diagnostic only; it does not include the atom of turning at the Eulerian corner.

1 The quantity and its normalisation

Use the continuous extension at zero in

$$r_E(\theta) = \frac{2\sin(\theta/2)}{\theta}, \qquad r_L(\theta) = b + a\cos\theta,$$

where

$$b = \frac{2}{\pi}\,\mathrm{Si}(\pi/2), \qquad a = \frac{2}{\pi}\big(\mathrm{Si}(3\pi/2) - \mathrm{Si}(\pi/2)\big).$$

Primes below denote differentiation with respect to θ. Define

$$v_i(\theta) = \sqrt{r_i(\theta)^2 + r_i'(\theta)^2}, \qquad k_i(\theta) = \frac{r_i^2 + 2(r_i')^2 - r_i r_i''}{(r_i^2 + (r_i')^2)^{3/2}}, \quad i \in \{E, L\}.$$

The signed numerator is positive on each smooth arc, so this formula equals the unsigned curvature used in the manuscript. For the Eulerian curve, it equals

$$\frac{3\theta\sin^2(\theta/2) + 2(\theta - \sin\theta)}{\theta^3} > 0 \quad (0 < \theta \le \pi),$$

with limiting value $13/12$ at zero. For the limaçon, its minimum is $(b-a)(b-2a) > 0$; the program verifies $b > 2a > 0$.

For definiteness, both full-arc normalised arclength parameters start at $\theta = -\pi$ and follow increasing θ to π. The profiles and speeds are even. On the positive half-arc put

$$x = \theta/\pi, \qquad H_i(x) = \pi v_i(\pi x), \qquad A_i(x) = \int_0^x H_i(t)\,\mathrm{d}t, \qquad \ell_i = A_i(1).$$

Let $x_i(v)$ be the unique solution of $A_i(x) = v\ell_i$, for $0 \le v \le 1$. Symmetry of the original full-arc integral gives exactly

$$D_\kappa^2 = \int_0^1 \left[k_E(\pi x_E(v)) - k_L(\pi x_L(v))\right]^2 \mathrm{d}v. \tag{2}$$

There is no omitted factor of two: the two half-arcs contribute equally and $v = 2u - 1$ on the second half of the original parameter interval.

Lemma 1 (Uniform inverse bound)**.** *For $0 \le x \le 1$, one has $H_E(x) \ge 2$ and $H_L(x) > 2$. Consequently, for every exact trial point $g \in [0, 1]$,*

$$|x_i(v) - g| \le \frac{|A_i(g) - v\ell_i|}{2}, \qquad |x_i(v) - x_i(c)| \le \frac{\ell_i}{2}|v - c|. \tag{3}$$

Proof. Concavity of sine on $[0, \pi/2]$ gives $r_E \ge 2/\pi$, so $H_E \ge \pi r_E \ge 2$. For the limaçon, $r_L \ge b - a > 0$. Outward-rounded evaluation verifies $\pi(b-a) > 2$, whence $H_L \ge \pi(b-a) > 2$. The functions A_i therefore increase strictly from 0 to ℓ_i. The first inequality follows from the mean value theorem; the second follows from $x_i'(v) = \ell_i / H_i(x_i(v))$. □

Newton iteration is used only to obtain an accurate trial point. The interval residual and Lemma 1, rather than the iteration's stopping rule, establish the inverse enclosure.

2 Enclosing the removable singularity and all derivatives

The numerical computation avoids cancellation in the sinc formula at zero. It uses the entire series

$$r_E(\theta) = \sum_{k=0}^{\infty} \frac{(-1)^k \theta^{2k}}{2^{2k}(2k+1)!}.$$

Let R_m retain $k = 0, \dots, m$, and set $n = 2m + 2$. For $|\theta| \le 4$ and $0 \le j < n$, define

$$q_j = \frac{4}{(n+2-j)(n+1-j)}.$$

When $q_j < 1$, absolute summation of the differentiated tail gives

$$\frac{|(r_E - R_m)^{(j)}(\theta)|}{j!} \le \frac{4^{n-j}}{2^n (n+1)(n-j)!\, j!} \frac{1}{1-q_j}. \tag{4}$$

Indeed, the ratio of successive absolute differentiated terms is at most

$$\frac{\theta^2}{4} \frac{n+1}{n+3} \frac{1}{(n+2-j)(n+1-j)} \le q_j,$$

and this upper bound decreases along the tail.

The implementation uses $m = 80$. It shifts the finite polynomial to the real point or interval under consideration and adds the bound (4) to each derivative coefficient. The coefficients for r_E' and r_E'' follow by differentiation, including the corresponding tail bounds. All evaluation centres are checked to satisfy $|\theta| < 4$.

Real ball arithmetic then encloses the Taylor coefficients of H_i and k_i under addition, multiplication, division and the positive square root. Every speed denominator is checked to have a strictly positive enclosure. For the limaçon, sine and cosine power-series operations supply the coefficients directly. The series truncation symbol alone is not used as an error estimate: (4) and the explicit quadrature remainders below provide the required bounds.

3 Validated arclength and outer quadrature

Fix an even order p and a dyadic partition of $[0,1]$ into N panels. The principal run uses $p = 32$, $N = 64$, and 256-bit midpoint precision.

Arclength integration. For a panel $[a, a+h]$, interval series evaluation over the whole panel encloses $H_i^{(p)}(\xi)/p!$. If its absolute value is bounded by B, then for $0 \le d \le h$,

$$A_i(a+d) - A_i(a) \in \sum_{j=0}^{p-1} \frac{H_i^{(j)}(a)}{j!}\frac{d^{j+1}}{j+1} + \left[-\frac{Bd^{p+1}}{p+1}, \frac{Bd^{p+1}}{p+1}\right]. \tag{5}$$

Accumulating these enclosures gives both $A_i(g)$ and ℓ_i. Coefficient rounding errors and all previous-panel errors remain in the balls.

Inverse derivatives. At a base point x_0, form the series

$$F_i(s) = \frac{A_i(x_0+s) - A_i(x_0)}{\ell_i}.$$

Its constant term is exactly zero and its linear coefficient is positive. Formal series reversion therefore gives the derivatives of $x_i(v_0+s) - x_0$, where $v_0 = A_i(x_0)/\ell_i$. Composing with the curvature series gives derivatives of $k_i(\pi x_i(v))$. For interval x_0, the same ball operations enclose these derivatives for every base point in the interval. The residual enclosure determines x_0 at the panel centre. The second bound in (3) covers all inverse images of the panel.

Outer integration. Write $f(v)$ for the squared difference in (2). On a symmetric panel $[c-h, c+h]$, let a_j enclose $f^{(j)}(c)/j!$, and let B_p bound $|f^{(p)}(v)|/p!$ throughout the panel. Taylor's theorem gives

$$\int_{c-h}^{c+h} f(v)\,\mathrm{d}v \in \sum_{\substack{0\le j<p \\ j \text{ even}}} a_j \frac{2h^{j+1}}{j+1} + \left[-\frac{2B_p h^{p+1}}{p+1}, \frac{2B_p h^{p+1}}{p+1}\right]. \tag{6}$$

The interval inverse series on the whole panel supplies B_p. Summing (6) encloses D_κ^2; the positive square root encloses D_κ.

Proposition 1 (Certification criterion)**.** *Under the checked positivity and domain conditions above, the interval returned by the supplied program contains the value in (2). If that interval lies strictly inside a decimal's nearest-rounding cell, the decimal is correctly rounded.*

Proof. The entire-series bound encloses the radial derivatives. Ball arithmetic propagates those inclusions through the speed and curvature formulae. Equation (5) encloses every arclength evaluation, Lemma 1 encloses every inverse value, and series reversion encloses its derivatives. Equation (6) includes the complete quadrature remainder. Summation and the monotone square root preserve inclusion. Strict containment in a rounding cell excludes either possible rounding boundary. □

4 Computed certificate and checks

The principal run returns the following enclosure for D_κ:

$$L = 0.203783759536049443826080629422606828470494451196624581353928,$$
$$U = 0.203783759536049443826080629422606828474246989273842126201317.$$

The machine-readable record contains the complete enclosure for the square, the two half-lengths, the numerical parameters and the exact program hash. The wider interval (1) follows immediately.

Run	Precision (bits)	Order p	Panels N
Principal certificate	256	32	64
Finer certificate	320	40	128

The finer enclosure is contained in the principal enclosure. Both certify the same 24-place value, and both also pass the program's 36-place rounding test. The certificate is the enclosure and its error argument, not agreement between runs.

An independent 60-decimal-place computation uses direct Gauss–Legendre quadrature and Newton inversion of separately integrated arclength functions. Increasing the outer quadrature order from 32 to 48 to 64 converges towards the certified value; its order-64 estimate lies inside the principal enclosure. This is supplementary numerical corroboration, not an interval certificate. An exploratory ordinary ODE solver did not attain its requested accuracy and was rejected as evidence.

The manuscript's existing decimal therefore needs no numerical change. Its accompanying accuracy statement can now cite validated arclength inversion and Taylor quadrature. The normalisation should explicitly identify the common starting angle and orientation.

5 Reproduction and scope

The primary dependency is `python-flint==0.9.0`, which supplies FLINT/Arb ball and power-series arithmetic [1, 2]. The optional independent check uses `mpmath==1.4.1`. The certificate was run with Python 3.12.14.

From the supplement directory:

```
python3 certify_dk.py --order 32 --panels 64 \
  --output D_K_CERTIFICATE_32_64.json
python3 certify_dk.py --bits 320 --order 40 --panels 128 \
  --output D_K_CERTIFICATE_40_128.json
python3 verify_certificate.py
```

The first two commands fail if the returned enclosure cannot certify the manuscript's 24-place value. The last command checks the recorded enclosures, their rounding-cell inclusion, the independent comparison, and the program hash.

This is a rigorous numerical certificate conditional on the documented floating-point ball-arithmetic implementation and its standard software/hardware assumptions; it is not a proof-assistant formalisation. It certifies the stated smooth-arc scalar diagnostic, not a global equality of curvature measures and not the entire manuscript. The numerical definitions and displayed values are unchanged in V423; its accompanying accuracy statements now refer to this certificate.

References

[1] F. Johansson, *Arb: Efficient Arbitrary-Precision Midpoint-Radius Interval Arithmetic*, arXiv:1611.02831. https://arxiv.org/abs/1611.02831. See also the current FLINT documentation: https://flintlib.org/doc/index_arb.html.

[2] *python-flint 0.9.0 documentation: arb and arb_series.* https://python-flint.readthedocs.io/en/latest/arb.html; https://python-flint.readthedocs.io/en/latest/arb_series.html.